

Mathematical Formulation of the Microgrid Dispatch Problem with Non-Linear Battery Degradation

Optimization Project Formulation

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1 Problem Definition

We are solving a Finite-Horizon Discrete-Time Optimal Control Problem. The goal is to determine the optimal power dispatch of a stationary battery storage system over a finite horizon to minimize the total operational cost. The cost comprises the net grid energy exchange, grid usage fees, and a non-linear penalty representing battery cycle degradation.

2 Sets and Indices

- $\mathcal{T} = \{1, 2, \dots, N\}$: The optimization horizon.
- $t \in \mathcal{T}$: A discrete time step index, where the duration of each step is Δt (e.g., $\Delta t = 1$ hour).
- $\mathcal{S} = \{1, 2, \dots, S\}$: The set of price scenarios, representing intraday price uncertainty relative to the day-ahead market.
- $s \in \mathcal{S}$: Scenario index.

3 Input Data (Parameters)

3.1 System Parameters

- E^{max} : Maximum energy capacity of the battery (kWh).
- E^{min} : Minimum allowable energy capacity to prevent deep depletion (kWh).
- P^{max} : Maximum allowable charge and discharge power (kW).
- η_c : Charging efficiency ($0 < \eta_c \leq 1$).
- η_d : Discharging efficiency ($0 < \eta_d \leq 1$).
- α : Degradation penalty coefficient (\$/kW²). Translates physical stress into economic cost.

- λ^{grid} : Grid usage fee for network access and transaction costs (\$/kWh).
- E_0 : Initial battery state of energy at $t = 0$ (kWh).

3.2 Time-Series Parameters

- $P_{load,t}$: Household power demand at time t (kW). Note: This is currently a point estimate from the ML prediction service.
- $P_{pv,t}$: Solar photovoltaic generation at time t (kW).
- λ_t^{DA} : Day-Ahead electricity price at time t (\$/kWh), fetched from ENTSO-E.
- $\epsilon_{t,s}$: Intraday price spread (stochastic noise) at time t under scenario s (\$/kWh), simulated via an AR(1)-GARCH(1,1) process to capture volatility clustering.
- $\lambda_{t,s}^{buy}$: Price of purchasing electricity from the grid at time t under scenario s (\$/kWh). Defined as $\lambda_{t,s}^{buy} = \lambda_t^{DA} + \epsilon_{t,s}$.
- $\lambda_{t,s}^{sell}$: Price of selling electricity to the grid at time t under scenario s (\$/kWh). Defined as $\lambda_{t,s}^{sell} = 0.9 \cdot \lambda_t^{DA} + \epsilon_{t,s}$, which incorporates the 10% transaction/selling discount.

4 Decision Variables

These are the continuous variables optimized by the Interior Point solver at each time step $t \in \mathcal{T}$:

- $P_{buy,t} \in [0, 20]$: Power imported from the grid (kW).
- $P_{sell,t} \in [0, 20]$: Power exported to the grid (kW).
- $P_{ch,t} \in \mathbb{R}_{\geq 0}$: Power sent to the battery for charging (kW).
- $P_{dis,t} \in \mathbb{R}_{\geq 0}$: Power drawn from the battery for discharging (kW).
- $E_t \in \mathbb{R}_{\geq 0}$: Battery State of Energy at the end of time step t (kWh).

5 Mathematical Formulation

5.1 Objective Function

The objective is to minimize the **expected total cost** $\mathbb{E}[J]$ over all price scenarios:

$$\min \mathbb{E}[J] = \frac{1}{S} \sum_{s \in \mathcal{S}} \sum_{t=1}^N \left[\left(\lambda_{t,s}^{buy} P_{buy,t} - \lambda_{t,s}^{sell} P_{sell,t} + \lambda^{grid} (P_{buy,t} + P_{sell,t}) \right) \Delta t + \alpha (P_{ch,t}^2 + P_{dis,t}^2) \right] \quad (1)$$

Note on Grid Usage Fees: The linear term $\lambda^{grid}(P_{buy,t} + P_{sell,t})$ represents real-world costs associated with using the electricity grid, such as network access fees or transaction

spreads. Mathematically, this term also acts as a penalty for simultaneous buying and selling: even if $\lambda^{buy} = \lambda^{sell}$, any non-zero flow through the grid incurs a cost, ensuring the solver selects the most efficient net energy exchange.

Note on Non-Linearity: The quadratic term $\alpha(P_{ch,t}^2 + P_{dis,t}^2)$ models the non-linear degradation proxy. In battery physics, I^2R heating and high C-rates accelerate component wear. Squaring the power mathematically penalizes sudden, aggressive power spikes, forcing the solver to prefer slow, steady charging profiles.

5.2 Constraints

1. Power Node Balance:

At every time step, the total power entering the household system must perfectly equal the power leaving or being consumed.

$$P_{pv,t} + P_{buy,t} + P_{dis,t} = P_{load,t} + P_{sell,t} + P_{ch,t} \quad \forall t \in \mathcal{T} \quad (2)$$

2. Battery Energy Dynamics:

The energy state at time t depends on the energy at $t - 1$ plus the net energy flowing in/out, adjusted for conversion inefficiencies.

$$E_t = E_{t-1} + \left(P_{ch,t}\eta_c - \frac{P_{dis,t}}{\eta_d} \right) \Delta t \quad \forall t \in \mathcal{T} \quad (3)$$

3. Operational Power Bounds:

The solver cannot command the battery to charge or discharge faster than its physical limits.

$$0 \leq P_{ch,t} \leq P^{max} \quad \forall t \in \mathcal{T} \quad (4)$$

$$0 \leq P_{dis,t} \leq P^{max} \quad \forall t \in \mathcal{T} \quad (5)$$

4. State of Energy Bounds:

The battery must not be overcharged or deeply discharged.

$$E^{min} \leq E_t \leq E^{max} \quad \forall t \in \mathcal{T} \quad (6)$$

5. Grid Interaction Bounds:

Power exchanges with the grid are limited by physical infrastructure (e.g., 20 kW fuse).

$$0 \leq P_{buy,t} \leq 20, \quad 0 \leq P_{sell,t} \leq 20 \quad \forall t \in \mathcal{T} \quad (7)$$

5.3 Consolidated Optimization Problem

Bringing the objective and all constraints together, the complete stochastic microgrid dispatch problem is formulated as follows:

$$\begin{aligned}
\min_{\mathbf{P}, \mathbf{E}} \quad & \frac{1}{S} \sum_{s \in \mathcal{S}} \sum_{t=1}^N \left[\left(\lambda_{t,s}^{buy} P_{buy,t} - \lambda_{t,s}^{sell} P_{sell,t} + \lambda^{grid} (P_{buy,t} + P_{sell,t}) \right) \Delta t + \alpha (P_{ch,t}^2 + P_{dis,t}^2) \right] \\
\text{s.t.} \quad & P_{pv,t} + P_{buy,t} + P_{dis,t} = P_{load,t} + P_{sell,t} + P_{ch,t} & \forall t \in \mathcal{T} \\
& E_t = E_{t-1} + \left(P_{ch,t} \eta_c - \frac{P_{dis,t}}{\eta_d} \right) \Delta t & \forall t \in \mathcal{T} \\
& 0 \leq P_{ch,t} \leq P^{max} & \forall t \in \mathcal{T} \\
& 0 \leq P_{dis,t} \leq P^{max} & \forall t \in \mathcal{T} \\
& E^{min} \leq E_t \leq E^{max} & \forall t \in \mathcal{T} \\
& 0 \leq P_{buy,t} \leq 20, \quad 0 \leq P_{sell,t} \leq 20 & \forall t \in \mathcal{T}
\end{aligned} \tag{8}$$

Where $\mathbf{P}$ and $\mathbf{E}$ represent the vectors of all power and energy decision variables over the horizon $\mathcal{T}$.

6 Algorithm Justification

Because the objective function contains a convex non-linear term (the squared battery power limits) subject to linear equality and inequality constraints, the problem constitutes a Non-Linear Programming (NLP) model. This requires a specialized Interior Point algorithm, such as IPOPT (Interior Point Optimizer), to reliably compute the global minimum.